FOREIGN KEY

1. What is a Foreign Key?

A **Foreign Key (FK)** is a **constraint** used in a relational database to **link two tables**.

A **foreign key** in one table refers to the **primary key** in another table.

Simple definition for students:

A foreign key ensures that the value entered in a column **already exists in another table**.

2. Why Do We Need a Foreign Key?

Foreign keys are used to maintain **Referential Integrity**.

Referential Integrity means:

- You **cannot insert invalid data**
- You **cannot delete important data accidentally**
- Relationships between tables remain **consistent**

Real-life example:

- A student must belong to a **valid department**
- An order must belong to a **valid customer**

3. Parent Table and Child Table

Term	Meaning
Parent Table	Table that has the **Primary Key**
Child Table	Table that has the **Foreign Key**

📌 **Rule:**

Foreign key is **always created in the child table**

4. Basic Example

Parent Table: Department

```
CREATE TABLE Department (
    DeptID INT PRIMARY KEY,
    DeptName VARCHAR(50)
);
```

Child Table: Student

```
CREATE TABLE Student (
    StudentID INT PRIMARY KEY,
    StudentName VARCHAR(50),
    DeptID INT,
    FOREIGN KEY (DeptID) REFERENCES Department(DeptID)
);
```

Explanation:

- Department.DeptID → **Primary Key**
- Student.DeptID → **Foreign Key**
- Student can only enter a DeptID **already present** in Department table

5. Foreign Key Syntax

FOREIGN KEY (child_column)
REFERENCES parent_table(parent_column);

6. Inserting Data – How Foreign Key Works

Insert into Parent Table first ✔

INSERT INTO Department VALUES (101, 'Computer Science');
INSERT INTO Department VALUES (102, 'Commerce');

Insert into Child Table ✔

INSERT INTO Student VALUES (1, 'Amit', 101);

Invalid Insert ❌

INSERT INTO Student VALUES (2, 'Ravi', 999);

🚫 **Error:**

DeptID 999 does not exist in Department table

➡ This is **referential integrity in action**

7. What is Cascading in Foreign Key?

Cascading controls what happens to child records when:

- Parent record is **deleted**
- Parent record is **updated**

Types of Cascading:

1. **ON DELETE CASCADE**
2. **ON UPDATE CASCADE**

8. ON DELETE CASCADE

Meaning:

If a record in the parent table is deleted,
related records in the child table are **automatically deleted**.

Syntax:

FOREIGN KEY (DeptID)
REFERENCES Department(DeptID)
ON DELETE CASCADE;

Example:

```
CREATE TABLE Student (
    StudentID INT PRIMARY KEY,
    StudentName VARCHAR(50),
    DeptID INT,
    FOREIGN KEY (DeptID)
    REFERENCES Department(DeptID)
    ON DELETE CASCADE
);
```

Effect:

If we delete:

```
DELETE FROM Department WHERE DeptID = 101;
```

✔ All students belonging to DeptID 101 are **automatically deleted**

9. ON UPDATE CASCADE

Meaning:

If a primary key value in the parent table is updated,
the foreign key value in the child table is **automatically updated**.

Syntax:

```
ON UPDATE CASCADE
```

Example:

```
FOREIGN KEY (DeptID)
REFERENCES Department(DeptID)
ON UPDATE CASCADE;
```

Effect:

If DeptID changes from 101 → 201,
all related student records update automatically.

10. Combining DELETE and UPDATE Cascade

```
FOREIGN KEY (DeptID)
REFERENCES Department(DeptID)
ON DELETE CASCADE
ON UPDATE CASCADE;
```

✔ Most commonly used in real databases

11. Other Foreign Key Actions

Action	Meaning	Support with Oracle 10 XE edition

CASCADE	Automatically affects child records	Yes
SET NULL	Sets foreign key value to NULL	Yes
SET DEFAULT	Sets default value	No
RESTRICT / NO ACTION	Prevents delete/update	No

SET NULL
Meaning: When the parent record is deleted or updated, the foreign key value in the child table is set to NULL.

Syntax
```
ON DELETE SET NULL
ON UPDATE SET NULL
```

Example
```
FOREIGN KEY (DeptID)
REFERENCES Department(DeptID)
ON DELETE SET NULL;
```

How It Works
Delete DeptID = 10 from Department
Student.DeptID becomes NULL

Important Condition
📌 The foreign key column must allow NULL values

12. Adding Foreign Key to Existing Table
```
ALTER TABLE Student
ADD CONSTRAINT fk_dept
FOREIGN KEY (DeptID)
REFERENCES Department(DeptID);
```

13. Dropping a Foreign Key
```
ALTER TABLE Student
DROP CONSTRAINT fk_dept;
```

14. Modifying a Foreign Key

📌 SQL does **NOT allow direct modification** of a foreign key.

Steps:

1. Drop the existing foreign key
2. Add a new foreign key with required options

Example

```
ALTER TABLE Student
DROP CONSTRAINT fk_dept;
ALTER TABLE Student
ADD CONSTRAINT fk_dept
FOREIGN KEY (DeptID)
REFERENCES Department(DeptID)
ON DELETE CASCADE;
```

15. Adding Foreign Key with SET NULL

```
ALTER TABLE Student
ADD CONSTRAINT fk_dept
FOREIGN KEY (DeptID)
REFERENCES Department(DeptID)
ON DELETE SET NULL;
```

📌 **Note:** Column must allow NULL

PRACTICE QUESTIONS

Lab Exercise 1: Foreign Key Constraint – Insertion Validation

Aim:

To understand how FOREIGN KEY constraints control valid data insertion and maintain referential integrity.

Question:

Write SQL commands in Oracle to perform the following operations:

1. Create a table **DEPARTMENT** with the following attributes:
   - DEPT_ID – NUMBER (Primary Key)
   - DEPT_NAME – VARCHAR2(30)
2. Create a table **EMPLOYEE** with the following attributes:
   - EMP_ID – NUMBER (Primary Key)
   - EMP_NAME – VARCHAR2(30)
   - DEPT_ID – NUMBER (Foreign Key referencing DEPARTMENT)
3. Insert records into the **DEPARTMENT** table.
4. Insert valid records into the **EMPLOYEE** table using an existing DEPT_ID.
5. Attempt to insert an EMPLOYEE record with a DEPT_ID that does not exist in the DEPARTMENT table and observe the error.

Lab Exercise 2: Foreign Key Constraint – Deletion and CASCADE Effect
Aim:
To study the effect of **ON DELETE CASCADE** when deleting records from a parent table.
Question:
Write SQL commands in Oracle to perform the following operations:

1. Create a table **CUSTOMER** with the following attributes:
    - CUST_ID – NUMBER (Primary Key)
    - CUST_NAME – VARCHAR2(30)
2. Create a table **ORDERS** with the following attributes:
    - ORDER_ID – NUMBER (Primary Key)
    - ORDER_DATE – DATE
    - CUST_ID – NUMBER (Foreign Key referencing CUSTOMER with ON DELETE CASCADE)
3. Insert records into both CUSTOMER and ORDERS tables.
4. Delete a record from the CUSTOMER table.
5. Display the ORDERS table and observe the automatic deletion of related records.

Lab Exercise 3: Foreign Key Constraint – Abandonment of Records using SET NULL
Aim:
To understand how **ON DELETE SET NULL** handles child records when a parent record is deleted.
Question:
Write SQL commands in Oracle to perform the following operations:

1. Create a table **PROJECT** with the following attributes:
    - PROJECT_ID – NUMBER (Primary Key)
    - PROJECT_NAME – VARCHAR2(30)
2. Create a table **EMPLOYEE_PROJECT** with the following attributes:
    - EMP_ID – NUMBER
    - PROJECT_ID – NUMBER (Foreign Key referencing PROJECT with ON DELETE SET NULL)
3. Ensure that the PROJECT_ID column allows NULL values.
4. Insert records into both tables.
5. Delete a record from the PROJECT table.
6. Display the EMPLOYEE_PROJECT table and observe that the PROJECT_ID is set to NULL instead of deleting the record.